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UNIT-2 : SMART MATERIALS

IMPORTANT: Syllabus: Piezoelectric materials, Electrostrictive materials, Magnetostrictive materials, Magnetoelectric materials, Magnetorheological and Electrorheological fluids, Shape memory materials.

2.1 Piezoelectric Materials

Definition: Piezoelectricity (Greek "piezein" = to press) is the ability of certain crystalline materials to generate an electric charge when mechanical stress is applied (direct effect) and to deform when an electric field is applied (converse effect).

- Discovered in 1880 by the Curie brothers (Pierre and Jacques Curie) in quartz; converse effect predicted by Lippmann (1881).
- Occurs only in crystals that have no centre of symmetry (non-centrosymmetric). Out of 32 crystal classes, 21 are non-centrosymmetric and 20 of them are piezoelectric.

Mechanism

- In an unstressed crystal the centres of positive and negative charges coincide → no net dipole.
- When stress is applied, the lattice distorts, the charge centres separate → dipole moment / polarisation → surface charge appears.
- Reversing the stress (tension ↔ compression) reverses the polarity of the voltage.

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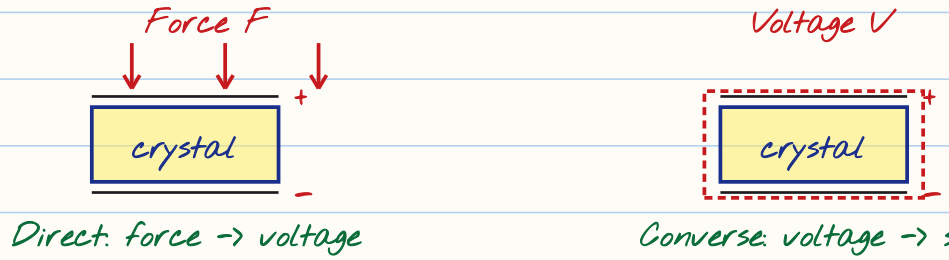


Fig: Direct and converse piezoelectric effect

Two effects

- Direct effect (sensor / generator mode) : mechanical stress ⟶ electric charge / voltage.
- Converse effect (actuator / motor mode) : electric field ⟶ mechanical strain / displacement.

Constitutive equations (linear piezoelectricity)

$$D = d\,T + \varepsilon\,E \quad \text{direct (sensor)}$$

$$S = s\,T + d\,E \quad \text{converse (actuator)}$$

- D = electric displacement (C/m2), T = stress (N/m2), S = strain, E = electric field (V/m), d = piezoelectric charge constant (C/N or m/V), s = elastic compliance, ε = permittivity.

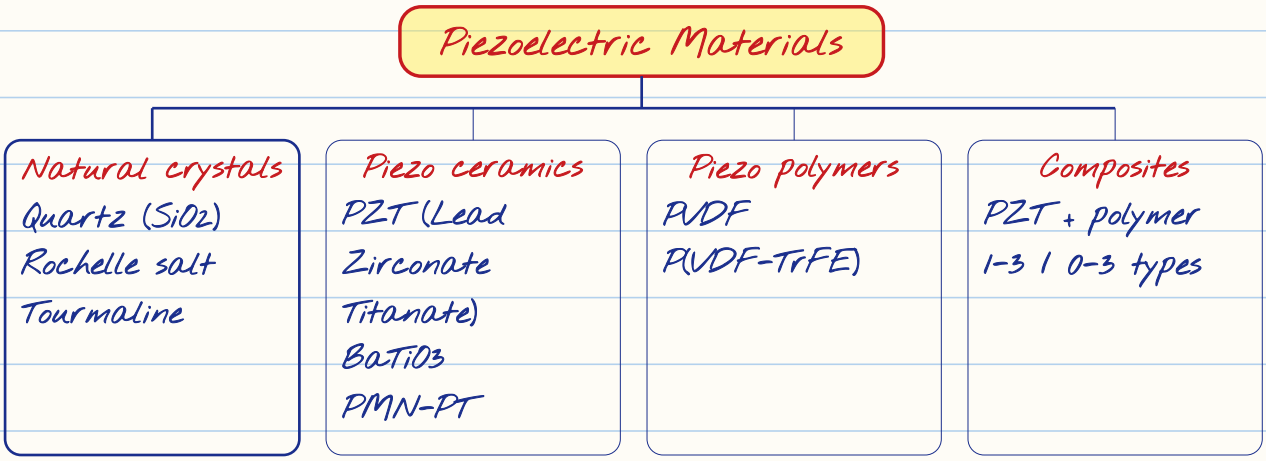
Important piezoelectric constants

- d (charge constant) : strain produced per unit field (m/V) or charge per unit force (C/N).
- g (voltage constant) : electric field produced per unit stress (Vm/N); $g = d / \varepsilon$.
- K (electromechanical coupling factor) : K squared = (stored mechanical energy) / (input electrical energy); measures the efficiency of energy conversion (PZT: $K \sim 0.7$).
- Mechanical quality factor Q_m and Curie temperature T_c.
- Subscript notation: d_{33} (field and strain along the polarisation axis - "3"), d_{31} (field along 3, strain perpendicular - "1"), d_{15}

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(shear).

Types of piezoelectric materials



- **Quartz** : very stable, high Q, used in oscillators, but small d (~2 pC/N).
- **PZT** ($Pb(Zr,Ti)O_3$): high d_{33} (~300–600 pC/N), high K; needs **poling** ; Curie temp ~ 200–350 C; contains lead.
- **PVDF** : flexible, light, low density, wide bandwidth, low d (~20 pC/N); used for thin-film sensors and hydrophones.

Poling

- Ferroelectric ceramics have randomly oriented domains → no net piezo effect.
- **Poling** = heating below Tc and applying a strong DC field (2–4 kV/mm) so that domains align, then cooling with field on. The ceramic now has a permanent polarisation.
- Heating above **Curie temperature** or applying a strong reverse field depolarises the material.

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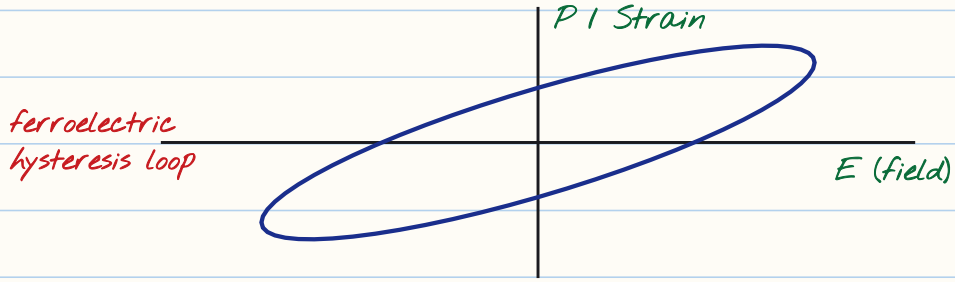


Fig: Polarization / strain hysteresis loop

- The strain - field curve of ferroelectric piezo ceramics shows hysteresis (~10-20%) and a butterfly loop at high field, which reduces accuracy of positioning.

Advantages and limitations

- Advantages: fast response (KHz - MHz), high force, precise (nm) displacement, no magnetic field, compact, no power at rest.
- Limitations: small strain (~0.1%), hysteresis and creep, brittle ceramics, needs high voltage, depoling at high temperature, lead toxicity.

Applications: Quartz oscillators and watches, ultrasonic transducers (medical imaging, NDT), microphones, sonar, accelerometers, piezo igniters (lighters), fuel injectors, inkjet printers, buzzers, energy harvesting, precision positioners (AFM / STM), ultrasonic motors.

2.2 Electrostrictive Materials

Definition: Electrostriction is the property of all dielectric materials by which they change shape (strain) when an electric field is applied, the strain being proportional to the square of the polarisation (or field).

$$S = Q P^2 = M E^2$$ quadratic effect

- Q = electrostrictive coefficient, P = polarisation, M = electrostrictive coefficient (field based). Strain is always

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positive expansion, independent of field direction.

Features

- Present in all dielectrics (even gases and liquids) but large only in some materials, e.g. relaxor ferroelectrics.
- Relaxor ceramics: PMN (lead magnesium niobate, $Pb(Mg_{1/3}Nb_{2/3})O_3$), PMN-PT, PLZT.
- No poling needed and almost no hysteresis (< 1–2%) when operated above the Curie / relaxor transition temperature.
- Very small ageing and creep; excellent repeatability; fast response.
- Strain is quadratic, so a bias field is applied to get linear actuation near the working point.
- Strongly temperature dependent: properties vary with temperature.
- No reverse (inverse) effect: electrostrictive materials do not produce voltage under stress (unless biased).

Comparison of piezoelectric and electrostrictive effects

Piezoelectric	Electrostrictive
Linear: $S = d E$	Quadratic: $S = M E^2$
Exists only in non-centrosymmetric crystals	Present in all dielectrics
Reverses sign with field direction	Same direction (expansion) for both polarities
Needs poling	No poling required
Hysteresis 10–20%	Hysteresis very low (~1%)
Direct and converse effects both exist	Only (practically) the field-strain effect

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Applications: Precision positioning stages, deformable mirrors in adaptive optics, micro-positioners, active vibration control, inKjet printing, sonar transducers, optical shutters.

2.3 Magnetostrictive Materials

Definition: Magnetostriction is the property of ferromagnetic materials by which they change their dimensions (length / shape) when magnetised in a magnetic field. The inverse effect (stress changes magnetisation) is the Villari effect .

- Discovered by James Joule (1842) in iron - the Joule effect. Strain is measured as lambda = delta L / L (parts per million).

Mechanism

- A ferromagnetic material consists of magnetic domains . When a field is applied, the domains rotate and the domain walls move → the lattice deforms → length change.
- Saturation magnetostriction (lambda s) is reached when all domains are aligned.

Effects

- Joule effect : field → change in length (actuator mode).
- Villari (inverse / magnetoelastic) effect : stress → change in magnetisation (sensor mode).
- Matteucci effect : twisting a magnetostrictive wire → induces axial magnetisation.
- Wiedemann effect : axial field + current → twist.
- Delta-E effect : Young's modulus changes with the magnetic state.

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Common magnetostrictive materials

Material	Magnetostriction (ppm)	Remarks
Nickel (Ni)	-35	Negative, old sonar material
Iron (Fe)	~ 14	Small, positive
Terfenol-D ($Tb_{0.3}Dy_{0.7}Fe_2$)	1000-2000	"Giant" magnetostriction; most used; brittle, costly
Galfenol (Fe-Ga)	~ 300	Machinable, rugged, low field
Metglas (Fe-Si-B amorphous)	~ 30	Soft, high permeability, sensors

- Terfenol-D = Terbium (TER) + Iron (FE) + Naval Ordnance Laboratory (NOL) + Dysprosium (D).
- Needs pre-stress (compressive, 7-10 MPa) and bias magnetic field for linear, large strain.

Definition: Magnetostrictive actuator = rod + drive coil (produces field) + permanent magnet (bias) + pre-load spring.

Advantages and limitations

- Advantages: high energy density, high force, large strain (~0.2%), fast, non-contact operation, low voltage but high current, high Curie temperature.
- Limitations: heavy and costly (rare earth), brittle, eddy current losses at high frequency, needs coil and bias magnet, hysteresis, temperature sensitivity.

Applications: Sonar transducers (underwater), active vibration control, ultrasonic cleaners, linear motors, fuel injectors, torque / force sensors, magnetostrictive delay lines, level sensors, energy harvesting.

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2.4 Magnetoelectric (ME) Materials

Definition: Magnetoelectric effect is the coupling between magnetic and electric properties of a material: an electric polarisation (P) is induced by a magnetic field (H) - direct ME effect, and a magnetisation (M) is induced by an electric field (E) - converse ME effect.

$P = \alpha H$ and $M = \alpha E$ alpha = ME coeff

- Predicted by Curie (1894), first observed in antiferromagnetic Cr_2O_3 (Astrov, 1960). Materials showing both ferroelectric and ferromagnetic ordering are called multiferroics.

Types of ME materials

- Single-phase materials: Cr_2O_3, $BiFeO_3$, $TbMnO_3$, $YMnO_3$ - ME effect weak and mostly at low temperature.
- Composite (two-phase) materials : a piezoelectric phase + magnetostrictive phase, ME effect much larger at room temperature.
  - Eg: $BaTiO_3$ - $CoFe_2O_4$, PZT - Terfenol-D, PZT - $NiFe_2O_4$, PVDF - Metglas laminates.

Product property (composite mechanism)

Magnetic field H → Magnetostrictive phase strains → Strain transferred to piezo phase → Voltage / polarisation produced

- ME effect in composite = (magnetic → mechanical) x (mechanical → electrical) : a product property - neither phase has it alone.
- Connectivity types: 0-3 (particulate), 2-2 (laminate), 1-3 (fibre / rod). Laminates give the highest coefficient.

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Note: ME voltage coefficient is expressed in mV/(cm.Oe); the effect is maximum at resonance (electromechanical resonance) and at a proper DC bias field.

Applications: Magnetic field sensors (very sensitive - pT level), current sensors, energy harvesters, transformers / gyrators, memory devices (MERAM), microwave devices, biomedical sensors.

2.5 Electrorheological (ER) and Magnetorheological (MR) Fluid

Definition: Rheological fluids are suspensions whose viscosity (flow resistance) changes rapidly and reversibly when an electric field (ER fluid) or magnetic field (MR fluid) is applied.

- ER fluid discovered by Willis Winslow (1947) - "Winslow effect"; MR fluid by Jacob Rabinow (1948).

Composition

Property	ER Fluid	MR Fluid
Particles	Dielectric (starch, silica, polymers, TiO2) 0.1-100 micron	Iron / carbonyl iron, ferromagnetic 1-10 micron
Carrier liquid	Silicone oil, mineral oil	Silicone / hydrocarbon oil, water
Additives	Surfactants, water (activator)	Surfactants, anti-settling agents
Applied field	Electric 1-5 KV/mm	Magnetic ~ 0.2-1 Tesla (low voltage 12-24 V)
Yield stress	2-5 KPa	50-100 KPa (10x more)
Response time	~ ms	~ ms

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Temperature range	Narrow	Wide (-40 to 150 C)
Sensitivity to impurity	High	Low

Working principle

- No field: particles are randomly dispersed → fluid behaves like a Newtonian liquid (low viscosity).
- Field ON: particles get polarised (dipoles) and line up as chains / fibril columns along the field direction within milliseconds → fluid turns into a semi-solid with yield stress.
- Field OFF: chains break → returns to liquid (reversible).

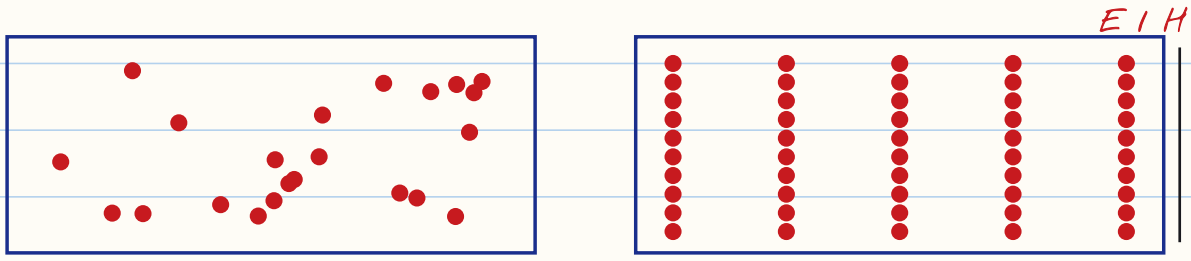


Fig: Particle alignment in MR / ER fluid

Bingham plastic model

$$\tau = \tau_y(E \text{ or } H) + \eta \times (\text{shear rate}) \qquad \tau > \tau_y$$

- τ_y = field dependent yield stress, η = plastic (off-state) viscosity. The fluid does not flow until applied shear stress exceeds τ_y.

Operating modes

- Valve (flow) mode - fluid flows between fixed poles (dampers, servo valves).
- Shear (direct shear) mode - fluid between moving plates (clutches, brakes).
- Squeeze mode - plates move towards each other (small-stroke,

high-force dampers).

Advantages / limitations

- Advantages: fast and reversible, simple, no moving parts (valve), controllable damping, semi-active → fail-safe.
- Limitations: particle sedimentation, abrasion, high-voltage need for ER, low yield stress of ER, in-use thickening, cost of MR fluid, temperature effects.

Applications: MR dampers in car suspensions (Audi, Cadillac), seismic dampers for buildings and bridges, washing-machine dampers, clutches and brakes, prosthetic knees, polishing of lenses, robot grippers; ER - clutches, engine mounts, haptic devices, valves.

2.6 Shape Memory Materials

Definition: Shape memory materials are materials which, after being deformed, can "remember" and return to their original shape when heated above a certain transformation temperature (or when a stimulus is applied).

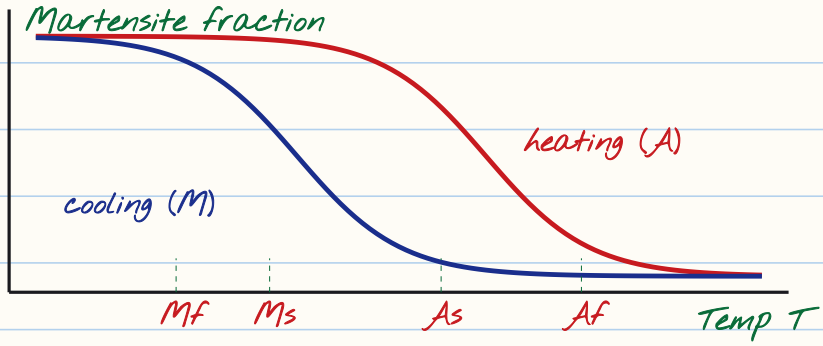
- Shape Memory Alloys (SMA): discovered by Arne Olander (1932, Au-Cd); NiTi (Nitinol = Nickel Titanium Naval Ordnance Laboratory) by Buehler (1962).

Phases of SMA

- Austenite (parent phase): high temperature, high symmetry (cubic B2), strong, stiff.
- Martensite (product phase): low temperature, low symmetry (monoclinic B19'), soft, easily deformed (detwinned). It can be twinned or detwinned.
- Transformation is diffusionless, solid-state, thermoelastic; characterised by four temperatures: Ms, Mf (martensite start /

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finish) and As, Af (austenite start / finish).

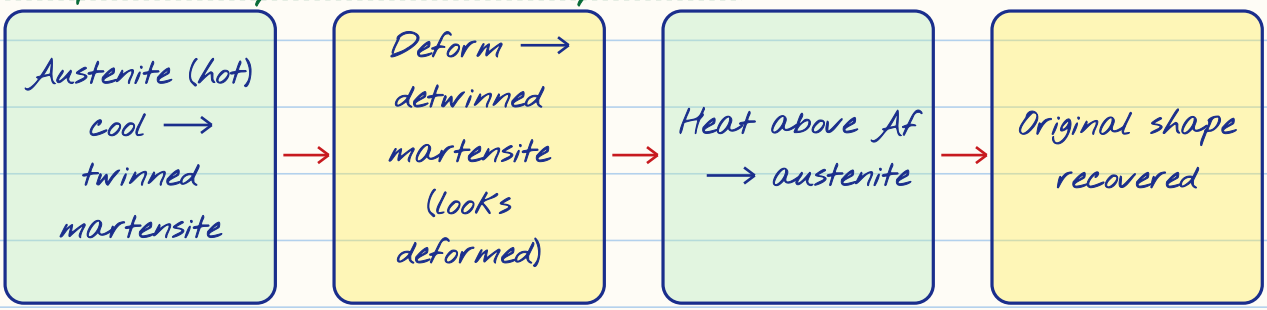


Ms - martensite start
Mf - martensite finish
As - austenite start
Af - austenite finish
Hysteresis: Af > Ms

Fig: Phase transformation of an SMA (hysteresis)

- The loop shows thermal hysteresis (Af - Ms ~ 20-50 C for NiTi).

Shape memory effect (one-way SME)



- One-way effect : remembers only the high-temperature shape; to reshape it must be deformed again at low temperature.
- Two-way effect : remembers both hot and cold shapes after "training" (thermo-mechanical cycling).

Pseudoelasticity (superelasticity)

- Above Af, a large stress induces stress-induced martensite ; on unloading it reverts to austenite and recovers up to 6-8 % strain (like a rubber spring, but metallic).
- Used in eyeglass frames, orthodontic wires, stents, guide wires, mobile phone antennas.

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Types of SMA

- NiTi (Nitinol) : best properties, 8% recovery strain, corrosion resistant, biocompatible; costly.
- Cu-based : Cu-Zn-Al, Cu-Al-Ni - cheaper, higher transformation temperature, brittle, less fatigue life.
- Fe-based : Fe-Mn-Si - cheap, low recovery.
- Magnetic SMA (ferromagnetic) : Ni-Mn-Ga - fast response to magnetic field (KHz), 6-10% strain.
- Shape memory polymers (SMP) : polyurethane, cross-linked polyethylene; triggered by heat / light, large strain, low stress.
- Shape memory ceramics and hybrids : Zirconia-based, SM composites.

Properties of SMA actuators

Property	NiTi value
Recovery strain	up to 8% (typically 4% in cycling)
Recovery stress	up to 800 MPa
Transformation temp	-100 to +100 C (tunable by composition)
Response time	Seconds (heating by current); slow cooling
Efficiency	< 5% (heat engine type)
Fatigue life	Depends on strain: 10^5 - 10^7 cycles

- Heated by Joule heating (passing current), cooled by air/water; bandwidth is limited by cooling time (~ 1-5 Hz).

Applications: Medical (stents, guidewires, orthodontic archwires, bone staples), aerospace (couplings, morphing wing, vibration damping), robotics (artificial muscles), thermostats, valves, fire sprinkler, eyeglass frames, couplings and fasteners, self-expanding structures in space (antenna).

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2.7 Quick Comparison of Smart Materials

Material	Input	Output	Features
Piezoelectric	Stress / Voltage	Voltage / Strain	Fast, linear, hysteresis
Electrostrictive	Electric field	Strain (quadratic)	Low hysteresis, no poling
Magnetostrictiv	Magnetic field	Strain	Large force, needs coil
Magnetoelectric	Magnetic field	Electric polarisation	Product property, composites
ER / MR fluid	Electric / Magnetic field	Viscosity / yield stress	Semi-active damping
SMA	Temperature / Stress	Shape recovery	Large force and strain, slow

2.8 Quick Revision / Important Questions

- 1. What is the piezoelectric effect? Explain direct and converse effects with equations and applications.
- 2. Differentiate piezoelectric and electrostrictive materials.
- 3. Explain magnetostriction. Write on Terfenol-D and its applications.
- 4. What is the magnetoelectric effect? Explain the composite ME material with example.
- 5. Explain the working principle of MR and ER fluids with a neat sketch. Write the Bingham model.
- 6. Explain shape memory effect and pseudoelasticity in SMA with the phase transformation diagram.

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IMPORTANT: Exam tip: for SMA always draw the hysteresis curve with Ms, Mf, As, Af; for MRIER draw the particle chains figure; for piezo draw the direct and converse effect.